

The information that should be contained in the HVAC&R system program includes

- design outdoor dry-bulb and wet-bulb temperatures (absolute and coincident);
- heating and cooling degree-days/hours;
- design wind velocity (and direction) for winter and summer;
- applicable zoning, building, mechanical, fire, and energy codes; and
- rate structure, capacity, and characteristics of available utilities and fuels.

Additional information regarding solar radiation availability and subsurface conditions would be included if use of a solar thermal system or ground-source heat pump was anticipated.

The environmental conditions to be maintained for each building space should be defined by

- dry-bulb and wet-bulb temperatures during daytime occupied hours, nighttime occupied hours, and unoccupied hours;
- ventilation and indoor air quality requirements;
- any special conditions, such as heavy internal equipment loads, unusual lighting requirements, noise- and-vibration-free areas, humidity limits, and redundancy for life safety and security; and
- acceptable range of conditions for each of the above.

An understanding of the functional use for each area is essential to select appropriate HVAC&R systems and suitable control approaches because the capabilities of proposed systems must be evaluated and compared to the indoor environmental requirements. For example, if some rooms in a building require humidity control while others do not, the HVAC system must be able to provide humidification to areas requiring it without detriment to the building enclosure or other spaces. Some areas may require cooling, while others need only ventilation or heating. This will affect selection of an appropriate system.

2.3.2 Design Phase

In conventional (business-as-usual) building projects, serious work on HVAC&R system design typically occurs in the later stages of the design phase. Projects where energy efficiency and/or green building design are part of the intent or building types where HVAC&R systems are absolutely integral to building design (labo-

tions, bed depth requirements, and maintenance procedures should be thoughtfully evaluated in light of the need for such a system.

Air washers (see Section 5.2.3) are also effective in reducing SO_2 and other acid-forming gases. But such scrubbers need continuous maintenance to keep the recirculated water from becoming highly corrosive and the reservoir a breeding ground for biological contaminants. Rare-book rooms and valuable artifact display or storage areas in museums and archival depositories are candidates for gas removal provisions, but the operations and maintenance staff should understand that careful maintenance is required for effective performance. Notwithstanding such gas removal provisions, the ventilation rate in these types of spaces should be maintained at no less than 15 cfm [7.1 L/s] per person (except as otherwise permitted by Standard 62.1) to limit the carbon dioxide concentration.

3.4 ROOM AIR DISTRIBUTION

3.4.1 Predicting Thermal Comfort

Standard 55 is based upon the presumption (explicitly stated in early versions of the standard) that “there is no minimum air movement that is necessary for thermal comfort,” although this is contrary to the experience of many building operators (see Section 3.4.2). The standard further prescribes a maximum rate of air movement (air speed) of 30 fpm [0.15 m/s] for winter and 50 fpm [0.25 m/s] for summer. Those values essentially agree with the data found in the “Space Air Diffusion” chapter in the *ASHRAE Handbook—Fundamentals*.

That chapter of the *Handbook* describes different methods of room air distribution. It also presents a space air distribution performance index (ADPI), which permits prediction of the comfort potential that can be achieved with a given supply air distribution design. The higher the index, the more uniform the conditions of comfort as determined by temperature variation from a control value and air velocities at various locations in the occupied zone. Using the guidance of that chapter, a designer can develop air distribution layouts that are likely to maintain a high probability of thermal comfort. Note that the ADPI concept is based upon research using a space (or control) temperature of 76°F [24.4°C]. Therefore, ADPI predictions apply only if the average space temperature is close to that value. For instance, if a room is maintained at 68°F

constant pressure drop. The *quick-opening* characteristic is usually applied where a process requires essentially ON/OFF operation. For example, a preheat coil in an outdoor air duct may be operated by a quick-opening valve, the controller of which has a sensor in the outdoor air.

A *linear* valve gives a uniform percentage change in flow for each percentage change in valve position. A typical application would be on a steam-heating coil operated by a controller whose sensor is on the discharge side of the coil. Since the coil output is uniform with steam flow, a linear valve provides a uniform change in temperature as the proportional controller increases or decreases the valve position to match the load to maintain a temperature setpoint.

An *equal-percentage* valve gives a very gradual change in flow for each percentage change in valve position near the closed position and a very rapid increase in flow near the fully open position. An example is the control of a hydronic heating or cooling coil. The nonlinear coil heat transfer characteristic (Figure 5-11a) is counteracted by the equal-percentage valve characteristic (Figure 5-11b), resulting in a nearly linear coil output. The flow-versus-valve-position characteristics (Figure 5-11c) assist the controller in matching the flow to the load and minimizing coil discharge fluctuations.

5.6.3 Valve Sizing

Sizing of control valves should be undertaken by determining the C_v (valve coefficient) required to provide the design flow conditions at full load, i.e., in the wide-open position. Manufacturers furnish values of C_v coefficients for each valve size. By definition, the C_v coefficient is the flow (in gpm) through a wide-open valve at a 1 psi pressure drop across the valve [or SI equivalents]. For water (with a specific gravity of 1), this reduces to

$$Q = C_v (\Delta p)^{0.5}, \quad (5-11)$$

where

- Q = flow rate, gpm (m^3/s);
- C_v = required valve coefficient; and
- Δp = pressure drop across valve, psi (Pa).

For steam, this formula becomes

$$W_s = 2.1 C_v [(p_1 + p_2) (p_1 - p_2)]^{0.5}, \quad (5-12)$$

where

- C_v = required valve coefficient;
- p_1 = valve inlet pressure, psia;

where the velocities are higher than in the ductwork. A very accurate pressure transducer is required with whatever scheme is selected.

Another flow-measuring technique uses hot-wire anemometers. If this approach is employed, appropriate air guides must be provided so that the anemometers measure only translational flow along the duct and not rotational or turbulent flow caused by local disturbances. The arrangement of properly connected hot-wire anemometers intended to provide average duct velocities must be specified in detail.

For most comfort applications, individual rooms can be controlled with proportional thermostats with local (in-the-room) setpoint adjustment. It is becoming common practice, however, to not provide for local thermostat adjustment in commercial/institutional buildings in an attempt to maximize energy efficiency and minimize maintenance. An alternative is to install temperature sensors with a slider bar that permits local adjustment. The slider bar can be programmed to establish how much temperature adjustment is allowed. The temperature control profile for the VAV dampers and heating coil (or baseboard heaters) should be as shown in Figures 10-7 and 10-8.

Room thermostats should not be located on exterior walls, in dead corners, or in other unsuitable locations where they are likely to be unduly affected by equipment, occupant, or solar radiation loads. The best locations are those that provide some air motion and allow thermostats to sense the average room temperature. The location of all thermostats should be indicated on the plans. If add-on, tamperproof covers are used, they should be constructed to allow free air movement over the thermostat. Metal or plastic box-type security enclosures with minimal air slits should not be used, since they do not provide enough airflow over the thermostat for it to respond rapidly to changing room conditions. If thermostats must be installed on exterior walls, insulating bases should be used and the wall penetration sealed airtight.

10.5.2 Single-Zone, Constant-Volume Air Systems

Very small single-zone air-handling units are thermodynamically equivalent to room fan-coil units or residential air-conditioning systems and can be controlled by simple two-position electric or electronic thermostat/valve systems common in this type of unit.

For larger single-zone systems, one of the schemes illustrated in Figures 10-9 and 10-11 can be used. Figure 10-9 shows a minimum configuration, which may provide inadequate control and should be used only on closely coupled systems, i.e., systems with